



NETWORK TRAFFIC ANALYSIS REPORT

Task Title: Network Traffic Analysis – Wireshark Live Capture

Name: Favour Chiemela

Tool Used: Wireshark

Capture Duration: Approximately 30–35 Minutes

Total Packets Captured: 88,959 Packets

1. Introduction:

As part of this task, I used Wireshark to capture and analyze live network traffic on my system. The aim was to observe how devices and applications communicate over a network and identify any interesting patterns that an investigator might notice during an investigation.

The traffic was captured while carrying out normal internet activities such as browsing websites and accessing online services. A total of 88,959 packets were captured during the session.

2. Findings and Observations:

Finding 1: DNS Requests to Online Services:

During the analysis, I observed approximately 320 DNS packets. DNS is responsible for translating website names into IP addresses so that devices can communicate with them.

Some of the domains observed included:

- www.youtube.com
- mail.google.com

This shows that before connecting to a website or service, the system first requested the address information needed to reach that service.

This helped to understand which online services a user is trying to access. And this shows that when later communications are encrypted, DNS requests can still provide useful clues about user activity.

```

DNS 93 Standard query 0x5e66 HTTPS rr4---sn-q4flrnld.googlevideo.com
DNS 93 Standard query 0xdfed A rr4---sn-q4flrnld.googlevideo.com
DNS 74 Standard query 0x5751 HTTPS www.google.com
DNS 74 Standard query 0xca58 A www.google.com
DNS 139 Standard query response 0xdfed A rr4---sn-q4flrnld.googlevideo.com CNAME r
DNS 202 Standard query response 0xca58 A www.google.com A 142.251.151.119 A 142.25
DNS 145 Standard query response 0x5e66 HTTPS rr4---sn-q4flrnld.googlevideo.com CNA
DNS 99 Standard query response 0x5751 HTTPS www.google.com HTTPS
DNS 71 Standard query 0x9029 HTTPS youtube.com
DNS 71 Standard query 0xba75 A youtube.com
DNS 87 Standard query response 0xba75 A youtube.com A 142.251.216.110
DNS 86 Standard query response 0x9029 HTTPS youtube.com HTTPS
DNS 75 Standard query 0xa197 HTTPS mail.google.com
DNS 75 Standard query 0xc95d A mail.google.com
DNS 125 Standard query response 0xa197 HTTPS mail.google.com SOA ns1.google.com
DNS 91 Standard query response 0xc95d A mail.google.com A 142.251.208.133
DNS 77 Standard query 0x7272 HTTPS www.google.com.ng
DNS 77 Standard query 0x37c2 A www.google.com.ng

s captured (1480 bits) on interface 0000 94 e9 79 5d c2 ab 5c 7d ae 86 1d 6a 08 00 45 00
teonTechno_5d:c2:ab (94:e9:79:5d:c2 0010 00 ab 39 a8 40 00 40 11 5b 39 c0 a8 12 01 c0 a8
e address (factory default) 0020 12 0f 00 35 d6 d9 00 97 f5 ca 75 20 81 80 00 01
ress (unicast) 0030 00 04 00 00 00 00 03 77 77 77 08 6d 73 66 74 6e
e address (factory default) 0040 63 73 69 03 63 6f 6d 00 00 01 00 01 c0 0c 00 05
ress (unicast) 0050 00 01 00 00 0c dc 00 20 03 77 77 77 08 6d 73 66
0060 74 6e 63 73 69 03 63 6f 6d 09 65 64 67 65 73 75
0070 69 74 65 03 6e 65 74 00 c0 2e 00 05 00 01 00 00
0080 00 f1 00 15 05 61 31 39 36 31 02 67 32 06 61 6b
0090 61 6d 61 69 03 6e 65 74 00 c0 5a 00 01 00 01 00
00a0 00 00 07 00 04 17 2d 99 e0 c0 5a 00 01 00 01 00
00b0 00 00 07 00 04 17 2d 99 cd
8.18.15

```

As an investigator, I checked the DNS list for unfamiliar or suspicious domains. If a device were infected with malware, we would see hidden DNS queries to strange, random website names that the user never typed. In this capture, every DNS query matched a safe website I actually visited.

Finding 2: High Volume of QUIC Traffic:

The capture contained approximately 37,407 QUIC packets, making it one of the most common protocols observed during the session.

QUIC is a modern protocol used by many websites and applications to make internet communication faster and smoother. During the analysis, I observed QUIC Initial packets, Handshake packets, and Protected Payload packets. Repeated communication with the same external addresses was also observed.

The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets, with a search filter 'quid' applied. The bottom pane shows the detailed view of packet 4487, which is a QUIC packet. The packet details include Ethernet II, Destination: LiteonTechno_5d:c2:ab (94:e9:79:5d:c2:ab), Source: zte_86:1d:6a (5c:7d:ae:86:1d:6a), Type: IPv4 (0x0800), Internet Protocol Version 4, User Datagram Protocol, and QUIC IETF. The packet length is 68 bytes, and it contains a Protected Payload (KP0) with a DCID of ef69076bbf0fb5d.

No.	Time	Source	Destination	Protocol	Length	Info
4474	52.935824500	192.168.18.15	142.251.209.206	QUIC	1285	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4475	52.936209100	192.168.18.15	142.251.209.206	QUIC	1201	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4478	53.097796700	142.251.209.206	192.168.18.15	QUIC	73	Protected Payload (KP0)
4479	53.126314100	192.168.18.15	142.251.209.206	QUIC	75	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4480	53.132499800	142.251.209.206	192.168.18.15	QUIC	1287	Protected Payload (KP0)
4481	53.133227200	142.251.209.206	192.168.18.15	QUIC	1292	Protected Payload (KP0)
4482	53.133242200	142.251.209.206	192.168.18.15	QUIC	825	Protected Payload (KP0)
4483	53.133244100	142.251.209.206	192.168.18.15	QUIC	238	Protected Payload (KP0)
4484	53.168176000	192.168.18.15	142.251.209.206	QUIC	80	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4487	53.325444200	142.251.209.206	192.168.18.15	QUIC	68	Protected Payload (KP0)
4599	81.480187500	192.168.18.15	142.251.209.206	QUIC	1285	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4600	81.480408600	192.168.18.15	142.251.209.206	QUIC	1292	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4601	81.480520100	192.168.18.15	142.251.209.206	QUIC	1292	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4602	81.480612200	192.168.18.15	142.251.209.206	QUIC	1292	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4603	81.480719000	192.168.18.15	142.251.209.206	QUIC	1292	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4604	81.480839300	192.168.18.15	142.251.209.206	QUIC	1109	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4605	81.705649000	192.168.18.15	142.251.209.206	QUIC	1285	Protected Payload (KP0), DCID=ef69076bbf0fb5d
4606	81.925336500	142.251.209.206	192.168.18.15	QUIC	73	Protected Payload (KP0)

Frame 4487: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface \Device\NPF{...} Ethernet II, Src: zte_86:1d:6a (5c:7d:ae:86:1d:6a), Dst: LiteonTechno_5d:c2:ab (94:e9:79:5d:c2:ab)
 Destination: LiteonTechno_5d:c2:ab (94:e9:79:5d:c2:ab)
 ..0. = LG bit: Globally unique address (factory default)
0. = IG bit: Individual address (unicast)
 Source: zte_86:1d:6a (5c:7d:ae:86:1d:6a)
 ..0. = LG bit: Globally unique address (factory default)
0. = IG bit: Individual address (unicast)
 Type: IPv4 (0x0800)
 [Stream index: 1]
 Internet Protocol Version 4, Src: 142.251.209.206, Dst: 192.168.18.15
 User Datagram Protocol, Src Port: 443, Dst Port: 55745
 QUIC IETF

QUIC packets made up over 40% of the traffic because I was actively using Google services like YouTube and Gmail during the capture. QUIC is Google's custom protocol designed to speed up video streaming and web loading. This volume is completely normal and matches my actual activity.

Finding 3: Encrypted TLS Communication:

TLS 1.2 and TLS 1.3 traffic were observed throughout the capture. Many of the packets appeared as Application Data, which means the information being exchanged was protected through encryption.

Although the actual contents of the communication could not be viewed, it was clear that secure communication was taking place between the device and online services and that traffic patterns can still be analyzed.

Destination	Protocol	Length	Info
142.251.209.206	QUIC	1292	Initial, DCID=af69076bbf0fb5da, PKN: 2,
216.58.223.238	TLSv1.2	879	Application Data
216.58.223.238	TLSv1.2	93	Application Data
216.58.223.238	TLSv1.2	910	Application Data
216.58.223.238	TLSv1.2	339	Application Data
216.58.223.238	TLSv1.2	311	Application Data
216.58.223.238	TLSv1.2	311	Application Data
216.58.223.238	TLSv1.2	311	Application Data
192.168.18.15	TLSv1.2	93	Application Data
192.168.18.15	QUIC	1292	Initial, SCID=ef69076bbf0fb5da, PKN: 3,
192.168.18.15	QUIC	1292	Initial, SCID=ef69076bbf0fb5da, PKN: 4,
192.168.18.15	TLSv1.2	124	Application Data
192.168.18.15	TLSv1.2	85	Application Data
192.168.18.15	TLSv1.2	93	Application Data
192.168.18.15	TLSv1.2	124	Application Data
192.168.18.15	TLSv1.2	85	Application Data
216.58.223.238	TLSv1.2	93	Application Data
192.168.18.15	TLSv1.2	124	Application Data

(656 bits), 82 bytes captured (656 bits) on interface Xde 0000 5c 7d a8 86 1d

As indicated, the presence of TLS confirms that communication was completely encrypted. This proves that even if an attacker intercepted these specific packets on the network, they would see nothing but unreadable information.

Finding 4: ARP Activity on the Local Network:

Approximately 129 ARP packets were observed during the capture.

Messages such as "Who has..." and "Is at..." were present. These messages help devices find and identify each other on the same network before communication begins.

[illegible]

3. Overall Analysis:

The network traffic captured during this exercise showed normal user activity and communication with online services. The analysis revealed frequent DNS requests, extensive use of encrypted communication through TLS, a significant amount of QUIC traffic, and normal ARP activity within the local network.

No obvious signs of malicious activity, unauthorized access attempts, or network attacks were observed during the capture period.

4. Conclusion:

This exercise helped me understand how devices communicate across a network and how Wireshark can be used to observe that communication. The analysis showed that many activities take place in the background whenever a user visits a website or uses an online service. It also demonstrated the importance of encryption in protecting information while it travels across the internet. Overall, this task provided practical experience in network traffic analysis and improved my understanding of how network communication works.